

FINAL PROJECT REPORT

Strategic Employee Attrition Intelligence Dashboard

Team ID: MCA | Date: 21 September 2026 | Power BI Analytics Project

1. Introduction

1.1 Project Overview

Employee attrition is a critical organizational challenge because the loss of employees directly affects workforce stability, productivity, and institutional knowledge. High turnover increases recruitment and training costs, disrupts team continuity, and can erode organizational performance. This project focuses on analyzing employee attrition using Microsoft Power BI to transform raw HR data into actionable workforce intelligence.

The dataset contains **5,000 employee records and 29 attributes** covering employee demographics, job details, compensation, workload, engagement, career growth, work arrangements, and attrition status. The analysis examines factors such as salary growth, overtime, burnout, engagement, work-life balance, manager support, performance rating, career level, department, and work mode to uncover attrition patterns and support data-driven retention decisions.

1.2 Objectives

- To analyze the overall employee attrition rate across the organization.
- To identify employee and workplace factors associated with attrition.
- To compare attrition across departments, job roles, work modes, and career levels.
- To analyze the relationship between workload, burnout, salary growth, and employee attrition.
- To develop an interactive Power BI dashboard for employee retention analysis.
- To provide meaningful, data-driven insights for workforce and retention strategy.

2. Project Initialization and Planning Phase

2.1 Problem Statement

The organization is facing a serious retention issue, with **895 out of 5,000 employees (17.9%) having left**. HR and leadership currently lack clear visibility into the primary drivers of turnover—whether compensation gaps, heavy workloads, burnout, poor engagement, limited career growth, or work-arrangement mismatches. Without this insight, the organization cannot prioritize its retention budget or design targeted interventions effectively.

From an HR Director perspective, the inability to understand why employees are leaving creates frustration and uncertainty about where to allocate retention resources. From a People Analytics

perspective, scattered data across systems makes it difficult to isolate operational drivers of turnover. Therefore, this project aims to build a **Strategic Employee Attrition Intelligence Dashboard** that presents attrition patterns through interactive Power BI visualizations, enabling data-driven decisions to reduce turnover and protect the workforce.

2.2 Proposed Solution

The proposed solution is an interactive Power BI dashboard that analyzes employee attrition using multiple employee and workplace factors. The dashboard converts raw employee data into meaningful visual information that can support workforce analysis. Key components include:

- KPI cards for critical employee metrics (Total Employees, Attrition Rate, Average Income, etc.).
- Charts for department- and job-role-level attrition comparison.
- Attrition analysis based on salary growth, overtime, burnout, and engagement.
- Work-mode and career-level comparative visuals.
- Interactive filters and slicers for multi-dimensional exploration.
- Dedicated report pages for detailed employee retention analysis.

2.3 Initial Project Planning

The project was organized into sequential phases to ensure systematic progress from problem definition through delivery:

Phase	Activity	Output
Problem Definition	Identify employee attrition problem	Problem Statement
Data Collection	Collect employee attrition dataset	Raw Dataset
Data Quality	Check data quality and consistency	Data Quality Report
Preprocessing	Clean and prepare data	Processed Dataset
Data Visualization	Create business questions and visuals	Power BI Visualizations
Dashboard Development	Design interactive dashboard	Power BI Dashboard
Report Development	Analyze dashboard findings	Final Report
Testing	Check filters, measures, and visuals	Performance Testing

Sprint-based delivery was planned as follows:

- Sprint 1 – Data Understanding & Preparation; Data Cleaning & Transformation; Performance & Career Growth Analysis; Dashboard Development & Testing (High priority user stories).

- Sprint 2 – Employee Attrition Analysis by department, job role, gender, and marital status (Medium priority).

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources

The Data Collection Plan focused on obtaining employee information related to compensation, workload, engagement, career progression, and work arrangement in order to identify factors associated with the 17.9% attrition rate. The primary raw data source is the provided **Employee Attrition CSV dataset** containing 5,000 employee records and 29 attributes.

Source	Description	Format	Records	Access
Employee Attrition Dataset	Demographic, job, compensation, workload, engagement, career growth, work arrangement & attrition data	CSV	5,000	Project Provided

3.2 Data Quality Report

Data quality assessment focused on accuracy and consistency of employee, compensation, workload, engagement, career progression, and attrition fields. Key checks and results:

Data Quality Check	Result
Total Records	5,000
Employees Left	895
Overall Attrition Rate	17.9%
Missing Values	No missing values identified
Duplicate Rows	No duplicate rows identified
Attrition Values	Yes / No (standardized)
Employee IDs	Unique

Additional quality considerations addressed during assessment included:

- Validation of engagement, burnout, and work-life balance score ranges for consistent comparison between retained and departed employees.
- Validation of compensation fields (monthly income, salary hike) and workload fields (overtime hours) before association analysis.
- Standardization of the Attrition field and creation of an Attrition Flag for reliable calculation of leavers and attrition rate.

3.3 Data Exploration and Preprocessing

The dataset was explored and prepared for Power BI analysis using Power Query. Preprocessing activities included data cleaning, data-type verification, category standardization, and creation of analytical groupings:

- Data cleaning and type verification for numerical and categorical fields (employee ID, age, income, overtime, scores, attrition).
- Standardization of categorical values (Department, Job Role, Work Mode, Gender, Marital Status, Education Level).
- Creation of analytical fields: Attrition Flag, Experience Group, Income Group, Workload Category, Engagement Category, Burnout Category, and Career Growth Indicator.
- Definition of key measures: Total Employees, Employees Who Left, Attrition Rate, Average Income, Average Engagement Score, Average Burnout Score, Average Overtime Hours, and Average Years Since Promotion.

4. Data Visualization

4.1 Framing Business Questions

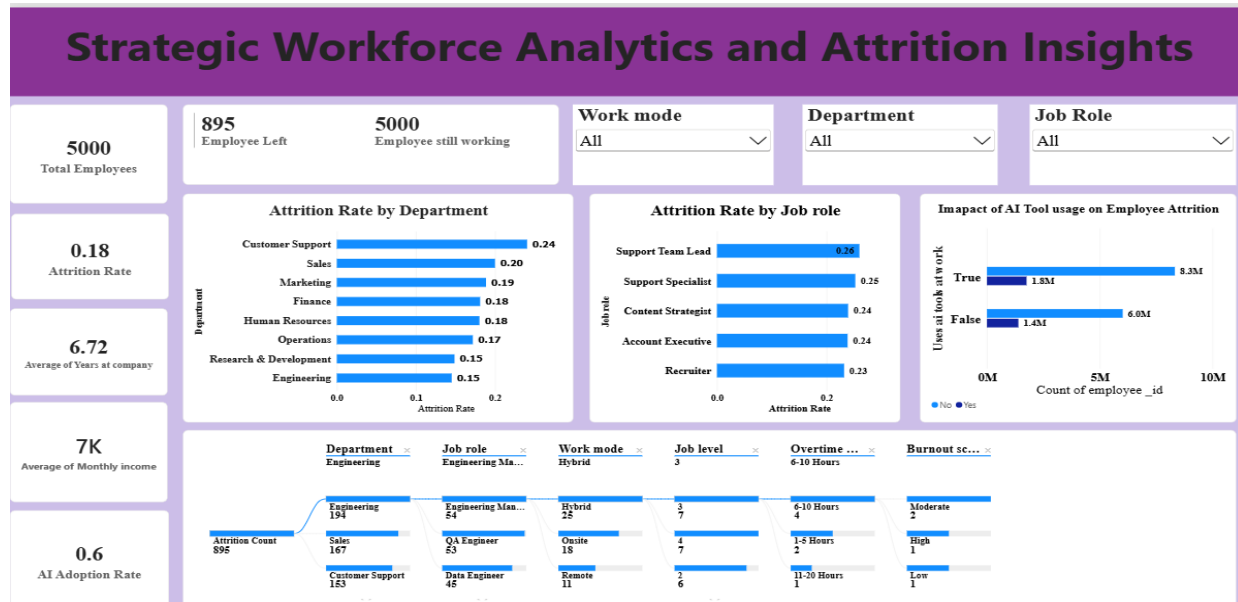
Visualization development was guided by specific business questions related to employee attrition. Appropriate Power BI visuals were mapped to each question to enable clear comparison of patterns:

No.	Business Question	Visualization
1	What is the overall employee attrition rate?	KPI Card + Donut Chart
2	Which departments have the highest attrition?	Clustered Column Chart
3	How does attrition vary across job roles?	Bar Chart
4	What is the relationship between engagement and attrition?	Column Chart / Scatter Plot
5	How does overtime workload affect attrition?	Column Chart
6	Does employee burnout influence attrition?	Scatter Plot / Column Chart
7	How does work mode affect attrition?	Donut / Column Chart
8	Does perceived AI job risk relate to attrition?	Scatter Plot / Column Chart

4.2 Developing Visualizations

Power BI was used to convert the employee dataset into interactive charts, KPI cards, and filters. These visualizations enable comparison of attrition patterns across employee and workplace characteristics, supporting clear and data-driven workforce analysis. Fields such as attrition, employee_id, department, job_role, engagement_score, overtime_hours_per_week, burnout_score, work_mode, and perceived_ai_job_risk were used to build the respective visuals.

5. Dashboard



5.1 Dashboard Design and Features

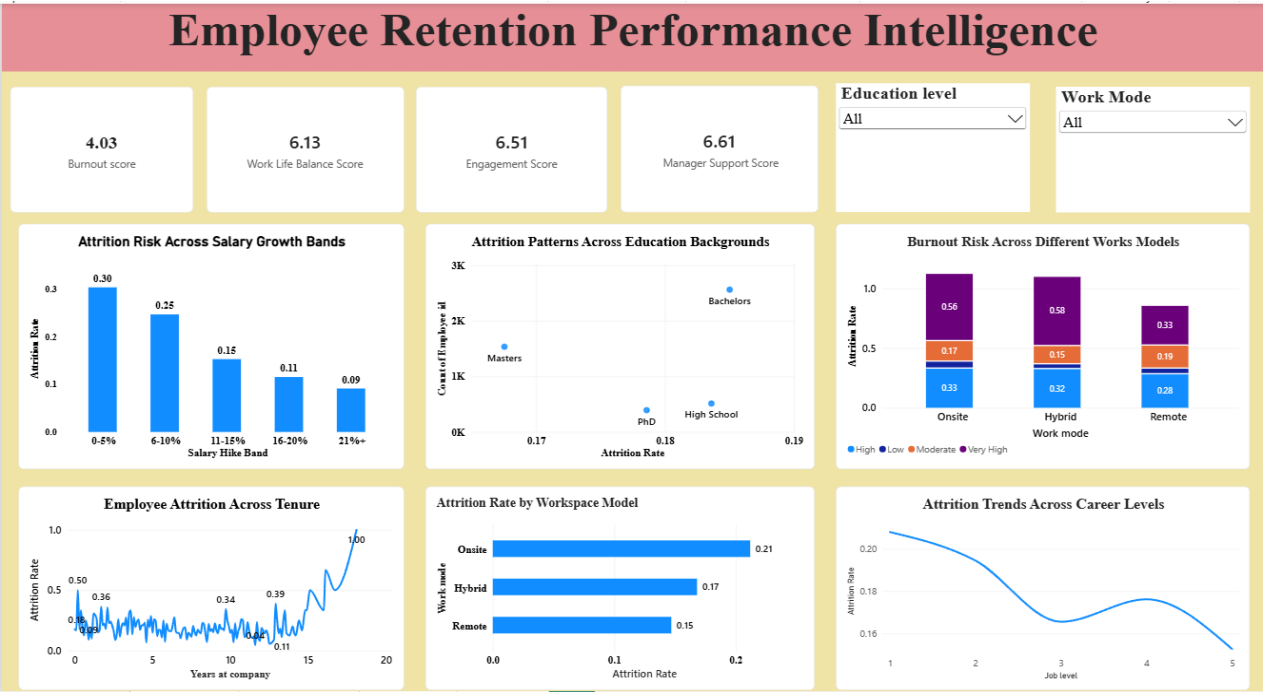
The Employee Attrition Dashboard was designed with a clear, organized layout using KPI cards, charts, interactive slicers, and a consistent professional theme. Design principles applied include a clear and intuitive layout, appropriate visualization choices, consistent colour theming, interactive filters, drill-down capability, and responsive arrangement of metrics.

- Clear layout with KPI metrics at the top and detailed visualizations below.
- KPI cards for Total Employees, Employees Left, Attrition Rate, Average Years at Company, Average Monthly Income, and AI Adoption Rate.
- Charts for department and job-role attrition comparisons.
- Interactive slicers for Work Mode, Department, and Job Role.
- Work-mode analysis and multi-category attrition views for flexible exploration.

Dashboard Insights

- Dataset covers **5,000 employees**, of which **895 have left** (approx. 18% attrition rate).
- **Customer Support** shows the highest department attrition at approximately **24%**, followed by Sales (20%) and Marketing (19%).
- **Support Team Lead** has the highest job-role attrition at approximately **26%**, followed by Support Specialist (25%).
- Average monthly income is approximately **7K**; average tenure is approximately 6.72 years.
- AI Adoption Rate displayed is **0.6**.

6. Report



The Power BI report analyzes employee retention and attrition using KPI cards, charts, and interactive filters. It focuses on factors including burnout, work-life balance, engagement, manager support, salary growth, education, work mode, tenure, overtime, performance rating, age, career level, and department. Interactive slicers such as Education Level, Work Mode, Gender, and Marital Status support detailed filtering. Key findings include:

- **Salary Growth:** The 0–5% salary-hike group has the highest attrition rate of 0.30; rates decline across higher growth categories, indicating salary growth as an important factor.
- **Work Mode:** Attrition rates are 0.21 (Onsite), 0.17 (Hybrid), and 0.15 (Remote), showing differences across working arrangements.
- **Overtime:** The 21–30 hours overtime group shows the highest attrition rate of 0.54, highlighting workload as a significant associate of turnover.
- **Performance Rating:** Attrition decreases from 0.29 (rating 2) to 0.07 (rating 5), showing variation across performance levels.
- **Burnout:** The Very High burnout category exhibits the highest attrition rate among all burnout levels.
- **Career Level & Department:** Attrition patterns vary across career levels 1–5 and across departments (overall rate 0.18).
- **Employee Metrics:** Avg. Burnout Score 4.03, Engagement 6.51, Manager Support 6.61, Age 37.12 years, Work Experience 11.70 years.

7. Performance Testing

7.1 Utilization of Data Filters

Interactive filters and slicers enable users to analyze specific employee groups. Filters include **Work Mode, Department, Job Role, Education Level, Gender, and Marital Status**. These allow multi-perspective exploration of attrition patterns and support targeted workforce analysis.

7.2 Calculation Columns / Measures

Key calculated measures supporting KPI cards and analytical visuals:

- Total Employees, Employees Who Left, Attrition Rate
- Average Income, Average Engagement Score, Average Burnout Score
- Average Overtime Hours, Average Years Since Promotion

7.3 Number of Visualizations

The Power BI project includes multiple visualization types used to present attrition patterns clearly:

- KPI Cards, Bar Charts, Column Charts, Donut Charts
- Scatter Plots, Line Charts, Slicers, and other interactive visuals

8. Conclusion / Observation

The **Strategic Employee Attrition Intelligence Dashboard** provides an interactive analysis of employee attrition using 5,000 employee records. Attrition patterns differ meaningfully across salary growth, work mode, overtime, performance rating, burnout, career level, department, and other employee characteristics.

The dashboard makes these patterns easier to understand through KPI cards, charts, and interactive filters, offering a solid foundation for further employee retention and workforce analysis. The findings represent **patterns and associations in the available dataset** and should not be interpreted as proof that any single factor directly causes attrition. They do, however, provide actionable direction for prioritizing retention efforts.

9. Future Scope

The project can be further improved by:

1. Adding historical employee data to analyze attrition trends over time.
2. Integrating additional HR datasets (recruitment, performance reviews, compensation, exit surveys).
3. Developing predictive analytics to identify employees at higher risk of attrition.
4. Adding detailed drill-through pages for departments and job roles.
5. Creating automated dashboard updates using regularly refreshed HR data.
6. Expanding analysis of AI usage and perceived AI job risk.
7. Adding advanced employee segmentation for targeted retention analysis.

10. Appendix

10.1 Source Code / Project Files

The project primarily uses **Power BI and Power Query** for data preparation, transformation, calculations, and visualization.

10.2 Project Dataset

Dataset: employee_attrition (3).csv — 5,000 employee records and 29 attributes used for the analysis.

10.3 Project GitHub URL Link

<https://github.com/poojithathumula13-rgb/Strategic-Employee-Attrition-Intelligence-Dashboard>